

Physics

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Topic: Newton's Laws of Motion

Subject: Physics

Key Concepts:

- **First Law (Law of Inertia):** An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.
- **Second Law (Law of Acceleration):** The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass ($F = ma$).
- **Third Law (Action-Reaction Law):** For every action, there is an equal and opposite reaction.

Explanation:

The **First Law** introduces the concept of inertia, stating that objects resist changes in their state of motion unless compelled by an external force.

The **Second Law** quantifies the relationship between force, mass, and acceleration, providing a mathematical framework for analyzing motion.

The **Third Law** emphasizes that forces always occur in pairs, acting on different objects but with equal magnitude and opposite direction.

Important Points:

- Formula: $F = ma$ (Force = mass $\times$ acceleration)

Summary:

Newton's Laws of Motion describe the relationship between forces, mass, and motion. The First Law defines inertia, the Second Law quantifies force and acceleration, and the Third Law explains the nature of interacting forces.

Key Terms:

- **Inertia:** The tendency of an object to resist changes in its state of motion.

- **Net Force:** The vector sum of all forces acting on an object.
- **Action-Reaction Forces:** Pairs of forces that are equal in magnitude and opposite in direction.